

Magnetism and Matter

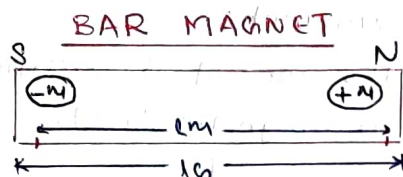
* Bar Magnet → $\begin{matrix} (-m) & (+m) \\ S & N \end{matrix}$

Magnetic pole → North pole (+m)
South pole (-m)

→ Same pole repel each other, opposite pole attract.

$$B = \frac{\mu_0}{4\pi} \frac{m}{r^2} \quad (\text{Magnetic field due to pole})$$

→ Magnetic field away from north pole & towards south pole.



$$\frac{l_m}{l_n} = 0.84$$

In 99% question

$$l_n = 1m$$

Magnetic Dipole Moment :-

$\vec{M} = m\vec{l}$

- Vector
- Direction S → N
- Unit = (Amp-m²)
- m → pole strength
- Unit = m = IA (Amp-metre)

(scalar) m (pole strength) ∝ Area
(vector) M (Magnetic moment) ∝ volume

Cutting of bar Magnet :-

m ∝ Area M ∝ volume

(1) cut along length (m/2), l

$$\begin{matrix} m/2 & m/2 \end{matrix} \quad \dots \quad M' = \frac{M}{2}, \quad m' = \frac{m}{2}$$

cut into n-equal part, the each part of:-

$$M' = \frac{M}{n}, \quad m' = \frac{m}{n}$$

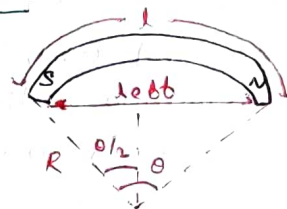
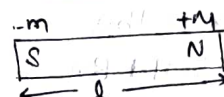
(2) cut ⊥ to length m, l/2

$$\begin{matrix} m/2 & m/2 \end{matrix} \quad M' = \frac{M}{2}, \quad m' = m$$

cut into n-equal part:-

$$M' = \frac{M}{n}, \quad m' = m$$

Bending of bar Magnet :-



$$M' = \frac{M \sin(\theta/2)}{(\theta/2)}$$

If semicircular Arc

$$\theta = \pi, \quad M' = \frac{2m}{\pi}$$

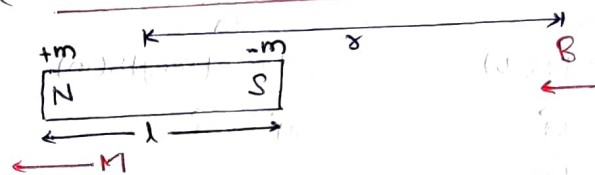
If complete circular arc

$$M' = 0, \quad \theta = 2\pi$$

n-turn → $M = IAN = ml$

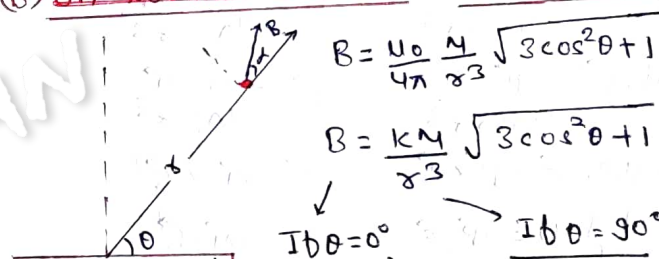
Magnetic field due to a Bar Magnet

(A) On Axial position



$$B_{axial} = \left(\frac{\mu_0}{4\pi} \right) \frac{2Mr}{(r^2 - l^2)^2} = \frac{2KM}{r^3}$$

(B) On Normal bisector :- At general point:-



$$B = \frac{\mu_0}{4\pi} \frac{M}{r^3} \sqrt{3\cos^2\theta + 1}$$

$$B = \frac{KM}{r^3} \sqrt{3\cos^2\theta + 1}$$

If $\theta = 0^\circ$
 $B_{axial} = \frac{2KM}{r^3}$

If $\theta = 90^\circ$
 $B = \frac{KM}{r^3}$

$$\tan\alpha = \frac{\tan\theta}{2}$$

Q → Find θ for which Mag. field ⊥ to direction of Magnetic moment??

Ans:- $\theta = \tan^{-1}(\sqrt{2})$

Magnetic Field lines

- (1) Magnetic field line is an imaginary curve, the tangent to which at any point gives direction of Magnetic field B at any point.
- (2) They form close loop.
- (3) They never intersect each other.
- (4) Outside → N to S
Inside → S to N
- (5) They come out & go into a magnetic material at any angle.

- ⑥ Larger the no. of field lines crossing per unit area, the stronger is the magnitude of the magnetic field B .
 B is larger around region (ii) than (i).

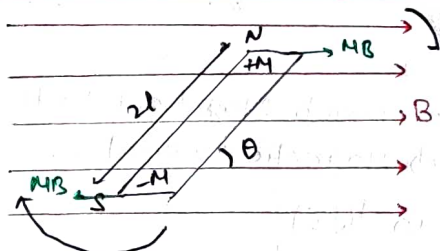


- ⑦ Electric field lines also called electric force line but magnetic field line does not called magnetic force line.

★ Analogy:-

Electrostatics	Magnetism
$\frac{1}{4\pi\epsilon_0}$	$\frac{\mu_0}{4\pi}$
Charge (q)	Pole strength (m)
$\vec{P} = q\vec{r}$	$\vec{M} = m\vec{r}$
$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$	$F = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^2}$
$\vec{F} = q\vec{E}$	$\vec{F} = m\vec{B}$
Axial, $\vec{E} = \frac{2\vec{P}}{4\pi\epsilon_0 r^3}$	$\vec{B} = \frac{\mu_0}{4\pi} \frac{2\vec{M}}{r^3}$
Equi, $\vec{E} = \frac{-\vec{P}}{4\pi\epsilon_0 r^3}$	$\vec{B} = \frac{-\mu_0}{4\pi} \frac{\vec{M}}{r^3}$
Torque, $\vec{\tau} = \vec{P} \times \vec{E}$	$\vec{\tau} = \vec{M} \times \vec{B}$
Potential Energy $U = -\vec{P} \cdot \vec{E}$	$U = -\vec{M} \cdot \vec{B}$

Dipole in uniform magnetic field



$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$\tau = MB \sin \theta$$

$$\theta = 0^\circ, \tau = 0$$

$$\theta = 90^\circ, \tau_{\max} = MB$$

$$\theta = 180^\circ$$

$$\tau = 0$$

$$U = -\vec{M} \cdot \vec{B}$$

$$U = -MB \cos \theta$$

$$\theta = 0^\circ, U = -MB$$

$$\theta = 90^\circ, U = 0$$

$$\theta = 180^\circ$$

$$U = MB$$

- Stable equilibrium at $\theta = 0^\circ$
- Unstable equilibrium at $\theta = 180^\circ$

Work done in rotating a Magnetic Dipole in uniform magnetic field:-

$$W_{\text{ext}} = \Delta U = U_f - U_i$$

$$W_B = -\Delta U = U_i - U_f$$

- Bar magnet will oscillate in uniform magnetic field about stable equilib.

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

I = Moment of Inertia

Vibrational Magnetometer:-

→ Experimental method to measure magnetic moment.

→ Based on the concept of oscillation of magnet in magnetic field.

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

★ Application -

- ① To find magnetic moment (M):-

$$M = \frac{4\pi^2 I}{T^2 B_H}$$

- ② To find ratio of magnetic moment of two magnet of same size:-

$$\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$$

- ③ To find ratio of mag. moment of two magnet of same size, oscillate one by one:-

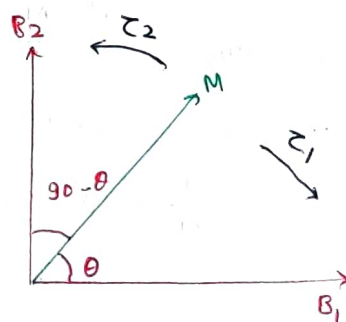
$$\frac{M_2}{M_1} = \left(\frac{T_1}{T_2} \right)^2$$

- ④ To find ratio of mag. moment of two bar magnet of unequal size:-

$$T_1 = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 + M_2)B}}, \quad T_2 = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 - M_2)B}}$$

Tangent law:-

$$\tan \theta = \frac{B_2}{B_1}$$



Magnetic field intensity (H):-

$$H = \frac{B_0}{\mu_0} = \frac{B_m}{\mu_m}$$

Magnetic permeability (μ):-

$$\mu = \frac{B}{H}$$

scales
Unit $\rightarrow$ Wb/Amp-m

Magnetic susceptibility (χ):-

$$\chi_m = \frac{I}{H}$$

- Scales
- No unit, no dimension

$\rightarrow$ A material with higher value χ_m can easily be magnetised.

Relation b/w μ_r & χ_m :-

$$\mu_r = 1 + \chi_m$$

Magnetic Material

$\rightarrow$ Not a magnet but can behave as magnet in external magnetic field. Eg - Iron

Atom:-

- $\rightarrow$ e⁻ revolve in circular path in atom.
- $\rightarrow$ current will be produced due to e⁻ motion.
- $\rightarrow$ that current will produce at the centre of circular path. (Mag. moment).

In paired e⁻ atom, two e⁻s are in opposite spin $\rightarrow$ **Diamagnetic**

$$\vec{M}_{\text{atom}} = 0$$

In unpaired e⁻ atom, $\rightarrow$ **Para/ferro**

$$\vec{M}_{\text{atom}} \neq 0$$

but $\vec{M}_{\text{crystal}} = 0 \rightarrow$ due to random orientation of atom

- $\rightarrow \mu_m > \mu_0$
- $\rightarrow \chi = +ve, \text{small}$

① Diamagnetism:-

- $\rightarrow$ Move from stronger to weak.
- $\rightarrow$ Universal property
- $\rightarrow$ Magnetise hota hai lenz law se (law of inertia).

$$\chi_m = -ve$$

$$\mu_m < \mu_0$$

$$\mu_r < 1$$

$$\mu_r < 1$$

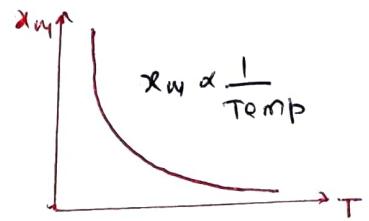
Independent of T

② Paramagnetism

$\rightarrow$ Tendency to move from weak magnetic field region to strong mag field.

$\rightarrow$ Magnetise hota hai lenz se.

$$\chi_m = +ve (\text{small})$$
$$B_{\text{net}} > B_0$$
$$\mu_m > \mu_0$$
$$\mu_r > 1$$

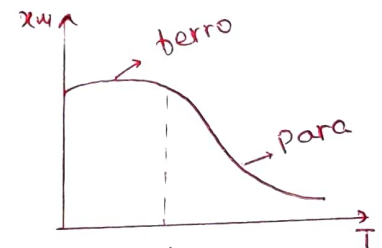


③ Ferromagnetism

$\rightarrow$ Strongly attracted to magnetic subst.

$\rightarrow$ Domain formation.

$$\chi_m = +ve$$
$$B_{\text{net}} \gg B_0$$
$$\mu_m \gg \mu_0$$
$$\mu_r \gg 1$$



$\rightarrow$ At higher temp, domain break.

$$\chi_{\text{para}} \ll \chi_{\text{ferro}}$$

Curie's law:- (Paramagnetic subst).

$$I = \frac{c B_0}{T}$$

$$\chi_m = \frac{c \mu_0}{T}$$

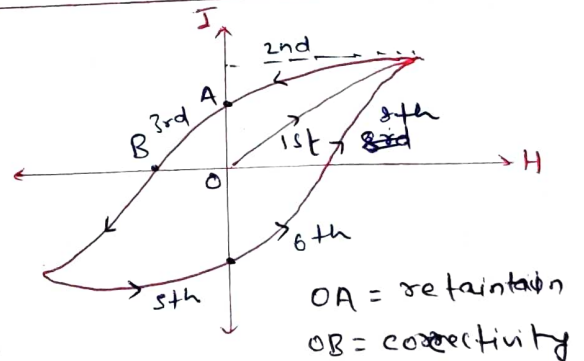
Curie Weiss law:-

$$\chi_m = \frac{c}{T - T_c}$$

I-H curve for ferromagnetic subst:-

I $\rightarrow$ Induced mag. moment per unit volume

Area of loop $\propto$ Energy loss



• ~~Paramagnet~~

• Permanent magnet

High retentivity, high coercivity

• Electro magnet

low retentivity, High permeability